LITERATURE REVIEW

LSTM-GRU Based Efficient Intrusion Detection in 6G-Enabled Metaverse Environments

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Domain: Cybersecurity / 6G Networks / Metaverse **Model Focus:** Hybrid LSTM–GRU **Year:** 2024

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1. Research Problem

The rapid growth of 6G networks and Metaverse technologies has introduced new cybersecurity challenges. Traditional intrusion detection systems struggle to identify sophisticated attacks and sequential network behavior accurately. Attack patterns in modern network environments are often sequential and time-dependent, making conventional machine learning approaches less effective.

To address this gap, the researchers proposed a hybrid deep learning model that combines Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks. The goal was to improve intrusion detection accuracy while reducing false alarms in complex, high-speed network environments.

2. Dataset

The study used a benchmark intrusion detection dataset containing both normal and malicious network traffic. The dataset includes multiple attack categories, enabling evaluation of the proposed model under diverse cybersecurity scenarios. Preprocessing and normalization steps were applied to prepare the sequential features for model training.

3. Methodology

The research workflow consisted of the following stages. First, the network traffic data was preprocessed and normalized. Sequential features were then extracted from the traffic flows and fed into a hybrid deep learning model combining LSTM and GRU layers. The trained model was evaluated using standard classification metrics including Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

4. Model Architecture

The proposed hybrid architecture consists of the following components:

- **Input Layer** – receives preprocessed sequential network features
- **LSTM Layer** – captures long-term sequential dependencies in traffic patterns
- **GRU Layer** – reduces computational complexity while retaining essential temporal information
- **Fully Connected Layer** – learns higher-level representations
- **Softmax Output Layer** – produces the final multi-class classification

In this design, the LSTM component specializes in modeling long-range temporal dependencies, while the GRU component improves training efficiency and reduces the overall computational cost of the network.

5. Results

The proposed hybrid LSTM–GRU model achieved approximately 90% classification accuracy. Compared with traditional intrusion detection approaches, the model demonstrated improved Precision, Recall, and F1-score. It also reduced the false positive rate and maintained strong performance across different attack categories, indicating robust generalization within the benchmark evaluation setting.

6. Limitations

- Evaluation was conducted primarily on benchmark datasets rather than live network traffic.
- The hybrid model requires significant computational resources for training.
- Real-time deployment in large-scale 6G environments requires further validation.
- Performance and feasibility on low-resource or edge devices remain uncertain.

7. Suggested Future Improvements

- Evaluate the model using real-world network traffic collected from operational environments.
- Apply transfer learning techniques to improve generalization across different network settings.
- Explore attention-based mechanisms and Transformer architectures for comparison.
- Optimize the model for edge computing and real-time intrusion detection deployment.

8. Critical Analysis

This paper demonstrates that combining LSTM and GRU provides a balanced solution for modern cybersecurity applications. LSTM effectively captures long-term dependencies in sequential network traffic, while GRU improves computational efficiency. The hybrid architecture achieves strong detection performance and lower false positives than many conventional methods.

However, the evaluation relies primarily on benchmark datasets, which may not fully reflect the complexity and variability of real-world network conditions. Future work should validate the approach on live traffic and systematically compare it with newer Transformer-based models. Overall, the study makes a valuable contribution by showing that sequential deep learning models remain highly effective for intrusion detection in emerging 6G and Metaverse environments.

— End of Literature Review —